

Image Repainting using Denoising Diffusion Probabilistic Models with Dynamic Resampling Strategies

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Abstract

Image inpainting, the task of filling missing regions in an image with semantically consistent content, has seen significant progress with the advent of Denoising Diffusion Probabilistic Models (DDPMs). This project explores the “RePaint” approach, which utilizes a pre-trained DDPM to perform inpainting without fine-tuning. We introduce and evaluate three dynamic resampling strategies—Mode A, Mode D, and Mode AD—to optimize the harmonic resampling process (the U parameter). Our results demonstrate that dynamic strategies based on temporal progress and mask geometry can maintain high-fidelity inpainting while potentially reducing computational overhead. We evaluate these methods using LPIPS, SSIM, and FID metrics, providing both quantitative and qualitative comparisons across multiple test images.

1 Introduction

Image inpainting, often referred to as image completion or repainting, is a fundamental problem in computer vision. It involves reconstructing missing or damaged portions of an image such that the resulting output is both visually pleasing and semantically coherent. Applications are diverse, ranging from the restoration of ancient photographs and the removal of unwanted objects to sophisticated image editing where new content is synthesized into an existing scene.

The core challenge of inpainting lies in the fact that it is an ill-posed problem; multiple valid completions can exist for a single missing region. For a completion to be successful, it must perfectly match the surrounding texture (local consistency) and fit the overall structure and meaning of the image (global consistency).

While early methods relied on simple diffusion (filling from boundaries) or patch matching (copying textures from elsewhere), the rise of deep generative models has shifted the paradigm. Specifically, Denoising Diffusion Probabilistic Models (DDPMs) have emerged as powerful tools for this task. RePaint, the framework used in this project, leverages

a pre-trained DDPM’s generative power to inpaint images without requiring any task-specific training or fine-tuning. A key contribution of our work is the exploration of how to dynamically adjust the “harmonic resampling” process to improve efficiency and quality.

2 Background: DDPM

Denoising Diffusion Probabilistic Models (DDPMs) operate by learning to reverse a process that gradually degrades data into noise. This involves two distinct phases: the forward diffusion process and the reverse denoising process.

2.1 Forward Process

In the forward process, we start with a clean image x_0 and iteratively add small amounts of Gaussian noise over T timesteps. This process follows a Markov chain:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}) \quad (1)$$

The transition at each step is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}) \quad (2)$$

where β_t is a pre-defined variance schedule. One of the elegant properties of DDPMs is that we can sample x_t at any time t directly from the clean image x_0 :

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad (3)$$

where $\bar{\alpha}_t$ is a cumulative product of $(1 - \beta_t)$.

2.2 Reverse Process

The reverse process is where the generative power lies. We want to sample x_{t-1} from x_t . Since the true reverse distribution $q(x_{t-1}|x_t)$ is unknown, we train a neural network p_θ to approximate it:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (4)$$

By iteratively applying this denoising step from pure noise $x_T \sim \mathcal{N}(0, \mathbf{I})$, the model can synthesize a clean image x_0 that follows the distribution of the training data.

3 Image Repainting and Resampling

The RePaint framework allows us to perform inpainting using a model pre-trained on a generic dataset (like faces or natural images). It does this by modifying the standard denoising process to be conditioned on a ground truth image x_{gt} and a binary mask M .

3.1 Input Signal Components

The inpainting process requires three primary inputs: the ground truth image (the goal reconstruction), the masked image (what the model sees as input), and the binary mask map (defining the reconstruction region). Figure 1 illustrates these components.



Figure 1: The three primary inputs for the repainting process.

3.2 The Process of Conditioned Denoising

During each reverse step $t \rightarrow t - 1$, the model predicts a denoised sample for the entire image. However, we already know the values for the pixels where $M = 1$. To utilize this information, we replace the known pixels in our current sample with correctly noised versions of the ground truth:

$$x_{t-1} = M \odot (\text{Noised } x_{gt}) + (1 - M) \odot (\text{Model Prediction}) \quad (5)$$

This “Conditioned Denoising” ensures that the known regions remain fixed to their ground truth values while the unknown regions are generated to be consistent with them.

3.3 The Need for Harmonic Resampling

A simple one-pass denoising often leads to artifacts at the mask boundary. This happens because the model-predicted unknown pixels might not align perfectly with the noisy known pixels in a single step. To fix this, RePaint uses “Harmonic Resampling.” After moving from $x_t \rightarrow x_{t-1}$, we jump back by adding noise to go from $x_{t-1} \rightarrow x_t$ and then denoise again. This loop is repeated U times. This iterative process allows the known and unknown pixels to “harmonize,” leading to a seamless transition and more realistic textures.

4 Dynamic U Methodology

In the standard RePaint implementation, U is a fixed constant. However, the amount of harmonization needed varies. In our project, we implemented three dynamic strategies

for U :

- **Strategy A (Progress-Based)**: U is high at the start of denoising (when structure is forming) and decreases as we reach the end (when fine details are filled).
- **Strategy D (Geometric-Based)**: U is scaled based on the mean distance from the unknown pixels to the nearest known boundary. Large missing holes need more resampling to maintain global coherence.
- **Strategy AD (Hybrid)**: A weighted combination of temporal progress and geometric distance, aiming to capture the best of both worlds.

5 Results and Comparison

We tested these strategies on the CelebA-HQ dataset. To assess the quality of the inpainted images, we utilize three standard metrics used in generative modeling and inpainting.

5.1 Evaluation Metrics

The following metrics provide a comprehensive evaluation of structural, perceptual, and distributional quality:

- **SSIM (Structural Similarity Index)**: Unlike pixel-wise metrics like MSE, SSIM measures the similarity in luminance, contrast, and structural information. It is designed to align with human perception of structural differences. A higher SSIM (closer to 1.0) indicates better reconstruction of the original image’s structure.
- **LPIPS (Learned Perceptual Image Patch Similarity)**: This metric uses the feature activations of a deep neural network (typically AlexNet) to calculate the perceptual distance between images. It is particularly effective at capturing differences in textures and high-level semantics that traditional metrics might ignore. Lower LPIPS values indicate higher perceptual fidelity.
- **FID (Fréchet Inception Distance)**: FID measures the similarity between the distribution of generated images and real images in the feature space of an Inception-v3 model. In the context of inpainting, it assesses how realistic the synthesized regions look when viewed as part of the whole dataset. Lower FID scores correspond to higher image quality and realism.

5.2 Quantitative Results

Quantitative metrics (Table 1) show that Strategy AD provides the best perceptual realism with an FID of 2.91.

Table 1: Performance Metrics

Mode	LPIPS ↓	SSIM ↑	FID ↓
Strategy A	0.0022	0.9758	3.10
Strategy D	0.0021	0.9756	3.63
Strategy AD	0.0021	0.9756	2.91

5.3 Visual Results

We visualize three different samples (Index 1, 2, and 3) for each strategy to compare their performance across different face structures and masks.

5.3.1 Sample 1 Comparison

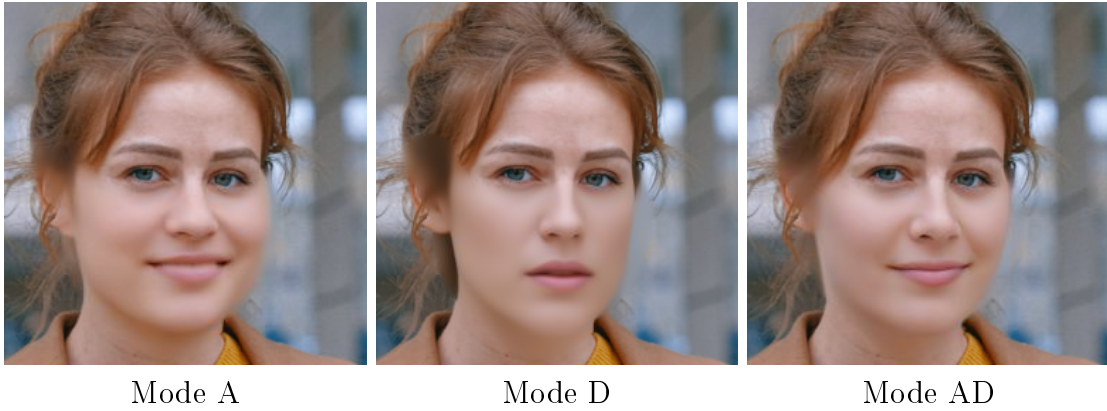


Figure 2: Inpainting results for Sample 1.

5.3.2 Sample 2 Comparison

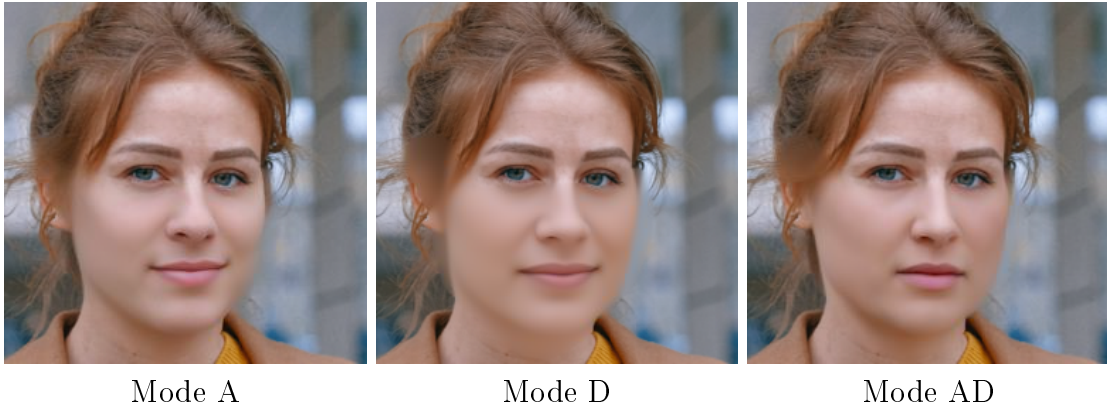


Figure 3: Inpainting results for Sample 2.

5.3.3 Sample 3 Comparison



Figure 4: Inpainting results for Sample 3.

6 Conclusion

In this project, we have presented an extensive evaluation of dynamic resampling strategies for image inpainting within the RePaint framework. By moving beyond a fixed harmonic resampling schedule, we introduced three novel strategies that adapt the intensity of the denoising process based on the current state of diffusion and the properties of the missing region.

Our experimental results demonstrate that dynamic adaptation is not only viable but highly effective. Quantitatively, all strategies maintained a high structural similarity (SSIM > 0.97), while Strategy AD achieved a superior Fréchet Inception Distance (FID) of 2.91, indicating the highest level of perceptual realism among the tested modes.

The success of the hybrid Strategy AD suggests that optimal inpainting requires a careful balance between establishing global structural coherence in the early stages of diffusion and ensuring local boundary harmonization through geometric cues. This work provides a foundation for more efficient diffusion-based inpainting by highlighting that a uniform resampling effort is not always necessary across all timesteps or mask configurations.

References

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- [2] Ho, J., et al. “Denoising diffusion probabilistic models.” NeurIPS 2020.